Mehul Kumar Jaiswal

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Professional Summary

Dedicated Computer Science student (B.Tech) specializing in backend logic, API architecture, and embedded systems. Experienced in translating foundational programming concepts into functional, real-world projects, ranging from memory-safe C databases to hardware-integrated IoT ecosystems. A proactive problem-solver and collaborative developer focused on writing efficient, scalable code and continuously expanding technical capabilities.

Skills & Achievements

- **Languages:** Python, Java, C/C++
- **Core Concepts:** Object-Oriented Programming (OOP), Data Structures & Algorithms
- **Applied Experience & Tools:** Kotlin, Flutter, Dart, Git, GitHub
- **Achievements:** 1st place - App Development Competition [App-E-Teaser] (JISTech 2K26),
  2nd Runner-Up - Theme Based Project [AI Enabled System] (JISTech 2K26)
  3rd Runner-Up - Crazy Build 2K26

Top Projects

Maargdarshan | AI-Enabled Smart Navigation Ecosystem — *Kotlin/Flutter, C/C++, Arduino, ESP32*

(Team Project - AI, App, Backend & Embedded System Co-Developer)

Live Prototypes: Website | App *(*Prototypes in active Beta testing)*

- **Embedded Systems & Hardware:** Engineered embedded C/C++ logic for a custom wearable IoT apparatus, integrating a multi-directional spatial sensor array and calibrating zero-latency localized haptic feedback.
- **Mobile Architecture:** Co-developed a persistent native mobile application designed to operate in a headless ambient state, serving as the primary voice-navigated AI router for the visually impaired user.
- **Telemetry & Emergency Routing:** Programmed the integration between the hardware node and mobile device, implementing live optical telemetry streaming and an OS-bypassing SOS dispatch system.
- **Backend Routing Server:** Engineered a robust Python API gateway to act as the ecosystem's central nervous system, managing real-time data flow, sensor telemetry routing, and bi-directional state synchronization between hardware nodes, Web Portal and Mobile application.

Chanakya-AI | Real-Time Competitive Intelligence Engine — *Python, FastAPI, Crawl4AI, REST APIs*

[Team Project - Co-Founder, Lead System Architect & Co-Backend Developer]

- Engineered the heavy backend infrastructure and scalable architecture for a real-time corporate strategy and competitive intelligence platform.
- Developed a complex API key rotation system and integrated the Crawl4AI engine to autonomously scrape, parse, and route live enterprise web data to the LLM pipeline.
- Optimized server-side execution and built an in-memory global context accumulator to bypass vector database overhead, ensuring zero-latency data delivery to the frontend.

Crop Management Database System — *C, Binary File Handling, RBAC, GCC*

[Solo Project - Developer & System Architect]

- **System Architecture & Security:** Architected a monolithic CLI-based AgriTech database featuring a custom 3-tier Role-Based Access Control (RBAC) system with secure credential verification and a fail-safe 3-attempt lockout mechanism.
- **Algorithmic File Management:** Engineered full CRUD functionality via binary file streams (`.dat`) and developed a Temporary Substitution algorithm to safely execute in-place record deletions and updates without database corruption.
- **Interoperability & Auditing:** Integrated a background module utilizing `<time.h>` for automated timestamped action logging, and built a data translation engine to export proprietary binary records into standard `.csv` format for spreadsheet analytics.
- **Runtime Stability:** Ensured zero-crash system reliability by implementing strict input validation and manual buffer-clearing memory routines to gracefully handle user edge cases and keystroke mismatches.

Education

B.Tech in Computer Science and Engineering (2025-2029)

JIS College of Engineering, Kalyani — Pursuing

Interests

- Participating in Hackathons and Beginner-Friendly Tech Events
- App Development & Cybersecurity
- Exploring Core Computer Science Concepts